

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** Authentication – Farmers can securely log in.
- **Non-Functional Requirements (NFR)** Performance – Prediction in less than 2 seconds.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Farmers can securely log in using username and password to access crop recommendations.</i>	High
2	Authorization levels	<i>Admin can manage datasets and models, while farmers can only view crop recommendations.</i>	Medium

3	External interfaces	<i>The system uses agricultural datasets stored in CSV files and machine learning libraries such as Scikit-learn and Pandas.</i>	High
4	Transactions processing	<i>The system receives soil and environmental parameters, processes them through the ML model, and returns suitable crop recommendations.</i>	High
5	Reporting	<i>The system displays predicted crops, analysis reports, and data visualizations to users.</i>	Medium
6	Business rules, etc.	<i>Crop recommendations are generated only when all required parameters are provided and validated.</i>	High
7	Compliance to laws or regulations	<i>Crop recommendations are generated only when all required parameters are provided and validated.</i>	Low
8	Other	<i>The system allows future integration with weather APIs and real-time agricultural data sources.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.NO	REQUIREMENT CATEGORY	Requirement Description	Priority (High/Medium/Low)
1	Performance & Speed	<i>The system should generate crop recommendations quickly</i>	<i>Prediction generated within 2 seconds.</i>
2	Scalability	<i>The system should support an increasing amount of agricultural data and users.</i>	<i>Supports more than 1000 users and large datasets.</i>
3	Security & Data Privacy	<i>User data and prediction records should be protected from unauthorized access.</i>	<i>Password protection and secure storage of records.)</i>
4	Reliability & Availability	<i>The application should work continuously without frequent failures.</i>	<i>99% system availability.</i>
5	Usability & Accessibility	<i>The interface should be simple and easy for farmers to use.</i>	<i>User-friendly interface with clear instructions.</i>

6	Other	<i>The system should be easy to update and run on different operating systems.</i>	Supports Windows, Linux, and macOS.
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